

CE 310 — Week 5 In-Class Exercise: The Bell Curve in Civil Engineering

Dataset: Week5_CE_Measurements.csv — 500 Field Observations

Topic: Normal Distribution · NORM.DIST · NORM.INV · Lognormal · ACI 318 Compliance

Before You Begin

Files needed:

- Week5_CE_Measurements.csv — 500 rows of field measurements combining three civil engineering datasets
- Wk05.04_In_Class_Answer_Sheet.xlsx — download from D2L; enter all answers here

Column map:

Col	Header	Description
A	RecordID	Row identifier (1–500)
B	Supplier	Concrete batch source: “Alpha” or “Beta”
C	TimeOfDay	Traffic observation window: “AM_Peak” or “PM_Peak”
D	fc_psi	Concrete compressive strength, psi
E	Speed_mph	Traffic spot speed, mph
F	CostPerCY	Construction cost, \$/CY

Answer Sheet instructions: On the Answers sheet, type the Excel formula or method you used in **Col C**, and the numerical answer in **Col E** of the row matching the question label — A1 in row A1, B3 in row B3. For ★ questions, also enter the intermediate value in **Col D**; leave Col D blank on every other question. **Col F** shows the expected unit and is not an input cell.

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761)'. Without the apostrophe Excel evaluates it and your method is lost. **Screenshots are not accepted;** the typed formula is what earns the approach credit.

Background

Concrete Compressive Strength and ACI 318

Concrete mix designs are evaluated against a specified compressive strength f'_c , set by the structural engineer of record. For this dataset, $f'_c = 4,000$ psi. ACI 318-19 (Section 26.12) requires that **no more than 10%** of test cylinder results fall below f'_c . A supplier whose batch exceeds this failure threshold must redesign its mix.

A supplier's mean strength matters — but so does variability. A high mean combined with a large standard deviation can still produce an unacceptable fraction of failures below f'_c . Section A asks you to quantify that risk for two suppliers with very different variability profiles.

Traffic Speed and the 85th Percentile

Transportation engineers use the **85th percentile speed** as the standard metric for evaluating posted speed limits. The 85th percentile is the speed at or below which 85% of drivers travel under free-flow conditions. Under MUTCD guidelines, a posted limit that differs substantially from the 85th percentile speed may not reflect actual operating conditions and warrants engineering review.

Section B separates observations into AM Peak and PM Peak periods. Congestion patterns, vehicle mix, and driver behavior often differ between those windows, producing meaningfully different speed distributions.

Lognormal Construction Costs

Construction costs per cubic yard are typically right-skewed: there is a practical lower bound and a few high bids that stretch the upper tail. When a variable X is always positive and $\ln(X)$ is approximately Normal, X follows a **lognormal distribution**. The lognormal model is parameterized by $\mu_{\ln}$ (mean of the log-transformed values) and $\sigma_{\ln}$ (standard deviation of the log-transformed values), computed on the natural log of the original data — not on the original values themselves.

Excel Quick Reference

Goal	Formula
Verify dataset row count	=COUNTA(A:A)-1
Conditional mean (by group)	=AVERAGEIF(B2:B501, "Alpha", D2:D501)
Conditional stdev (Excel 365)	=STDEV(FILTER(D2:D501, B2:B501="Alpha"))
Conditional stdev (array formula)	=STDEV(IF(B2:B501="Alpha",D2:D501)) — Ctrl+Shift+Enter
Count records in a group	=COUNTIF(B2:B501, "Alpha")
Normal CDF — $P(X \leq x)$	=NORM.DIST(x, μ , σ , TRUE)
Normal exceedance — $P(X > x)$	=(1 - NORM.DIST(x, μ , σ , TRUE)) * 100
Normal percentile — inverse CDF	=NORM.INV(probability, μ , σ)
Standard Normal z-score at percentile	=NORM.S.INV(probability)
Natural log (for lognormal setup)	=LN(value)
Lognormal CDF — $P(X \leq x)$	=LOGNORM.DIST(x, $\mu_{\ln}$, $\sigma_{\ln}$, TRUE)
Lognormal exceedance	=(1 - LOGNORM.DIST(x, $\mu_{\ln}$, $\sigma_{\ln}$, TRUE)) * 100

Setup (~5 min)

You do all of this week's work in **one file: your Answer Sheet**. The data, your scratch work, and your charts all live there, so everything you built is in the file you hand in. Add as many sheets as you like — only Instructions, Answers, Work, Charts and Written have to keep their names.

1. **Import Week5_CE_Measurements.csv** onto a sheet named Data that you add to your Answer Sheet:

- Data → Get Data → From Text/CSV → Load.

2. Confirm **500 rows** of data: `=COUNTA(A:A)-1`
 3. Confirm the column headers in row 1: RecordID, Supplier, TimeOfDay, fc_psi, Speed_mph, CostPerCY
 4. **Add a helper column on your Data sheet.** It goes beside the data because it holds one value per row. Label the first empty column LN_Cost — column G if your import started in A1 — and in the cell below the header enter:
`=LN(F2)`
 Fill this formula down through G501. You will use this column for the lognormal cost analysis in B17–B18.
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Section A — Concrete Compressive Strength (16 pts)

Section A fits a Normal model to concrete strength and uses it to answer the compliance question ACI 318 actually asks: what fraction of cylinders fall below the specified strength?

All Section A calculations use the fc_psi column (D2:D501). Where noted, use AVERAGEIF, COUNTIF, or FILTER to isolate observations by Supplier.

Rounding: psi values (A1, A2, A3, A7, A9) → nearest whole psi; skewness (A4) → 2 decimal places; probabilities (A5, A6) → 1 decimal place.

A1 (1 pt) — Mean Compressive Strength

Compute the average compressive strength across all 500 specimens.

A2 (1 pt) — Standard Deviation of Compressive Strength

Compute the sample standard deviation of fc_psi for all 500 specimens.

Enter the stdev in **Col E**. As a check, confirm `=COUNTA(D:D)-1` returns 500 — that count is not a graded entry on this question.

A3 (1 pt) — Median Compressive Strength

The median is the 50th percentile — the value that splits the ranked data in half.

When the median is close to the mean ($A3 \approx A1$), the distribution is approximately symmetric. A large gap between mean and median signals skewness.

A4 (1 pt) — Skewness of Compressive Strength

Compute the skewness of fc_psi for all 500 specimens.

A SKEW value between -0.5 and $+0.5$ is considered near-symmetric for engineering purposes. A Normal distribution has $SKEW = 0$. Compare your result to that benchmark.

Normal model check (ungraded): $n = 500$ and $|SKEW| < 0.5$, so the two conditions for `NORM.DIST` hold. Before A5, work out $z = (4,000 - A1) / A2$ — a large negative z means failures should be rare.

A5 (2 pts) — $P(f_c < 4,000 \text{ psi})$ for All Specimens

The Normal CDF gives the probability that a randomly selected specimen falls below a threshold. Using the mean from A1 and the standard deviation from A2, compute the probability that a specimen falls below 4,000 psi.

ACI 318 allows at most 10% below $f'_c = 4,000 \text{ psi}$. How does the combined dataset compare?

A6 (2 pts) — $P(4,200 < f_c < 5,000 \text{ psi})$ for All Specimens

The probability of falling in a range equals the CDF at the upper limit minus the CDF at the lower limit. Compute this range probability using the mean from A1 and standard deviation from A2.

68-95-99.7 check (ungraded): about 95% of specimens sit within $A1 \pm 2 \cdot A2$. Does the 4,200–5,000 psi band fall inside that, or out in a tail?

A7 (3 pts) — 10th Percentile Strength for Supplier Alpha

ACI 318 uses the 10th percentile of the fitted Normal distribution to assess compliance. Compute the 10th percentile for Supplier Alpha cylinders using their conditional mean and standard deviation.

The Work sheet of your Answer Sheet has a two-row supplier block — Alpha and Beta, at rows 7 and 8. Put Alpha's conditional mean in B7 and its conditional standard deviation in C7, then apply NORM.INV at the 10th percentile to those two cells. A9 does the same for Beta on row 8, and Chart 3 plots the last column of the same block.

A8 (MC · 1 pt) — ACI 318 Compliance: Supplier Alpha

Based on your result for A7, which statement about Supplier Alpha is correct?

- (A) Alpha is non-compliant — the 10th percentile falls below 4,000 psi
- (B) Alpha is compliant — the 10th percentile exceeds 4,000 psi
- (C) Compliance depends only on the mean strength, not the standard deviation
- (D) Alpha must achieve exactly 0% of cylinders below f'_c to be compliant

A9 (3 pts) — 10th Percentile Strength for Supplier Beta

Repeat the procedure from A7 for Supplier Beta: compute the conditional mean and standard deviation for Beta observations, then apply NORM.INV at the 10th percentile.

A10 (MC · 1 pt) — Why Supplier Beta Fails ACI 318

Based on your results from A7 and A9, why does Supplier Beta fail the ACI 318 compliance criterion?

- (A) Beta's mean strength is too low for any mix design to pass the spec
 - (B) Beta's standard deviation is large enough that, even with a mean above 4,000 psi, more than 10% of cylinders fall below f'_c
 - (C) ACI 318 uses the 5th percentile, not the 10th, for compliance evaluation
 - (D) Beta's mean strength is below 4,500 psi, which is the ACI minimum
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Section B — Traffic Speed and Construction Cost (60 pts)

Section B carries the same model into traffic speed and construction cost — including one variable that is not Normal at all — and shows what has to change when the distribution does.

Questions B1–B16 use the Speed_mph column (E2:E501). Questions B17–B18 use CostPerCY (F2:F501) and the LN_Cost helper column (G2:G501) you created in Setup.

Rounding: speeds and probabilities → 1 decimal place; μ_{ln} (B17) → 2 decimal places.

B1 (2 pts) — Mean Traffic Speed

Compute the average spot speed across all 500 observations.

B2 ★ (3 pts) — Standard Deviation of Traffic Speed

Compute the sample standard deviation of Speed_mph for all 500 observations.

Col E: stdev in mph · Col D: =COUNTA(E:E)–1 (should equal 500)

B3 (2 pts) — Median Traffic Speed

Compute the median spot speed across all 500 observations.

B4 (2 pts) — Skewness of Traffic Speed

Compute the skewness of Speed_mph for all 500 observations.

Compare B4 to A4. Both should be near zero — Normal distributions arise naturally in quantities governed by many small, additive effects.

B5 (3 pts) — $P(\text{Speed} \leq 45 \text{ mph})$

The posted speed limit on this corridor is 45 mph. What fraction of drivers are at or below the limit? Use the overall mean (B1) and standard deviation (B2) with NORM.DIST.

B6 (3 pts) — $P(\text{Speed} > 45 \text{ mph})$

What fraction of drivers **exceed** the posted limit? The exceedance probability is the complement of the CDF. Use the overall mean (B1) and standard deviation (B2).

Confirm that $B5 + B6 = 100\%$. If they do not, check for a missing parenthesis.

B7 ★ (4 pts) — 85th Percentile Speed (All Observations)

The 85th percentile speed is the standard traffic engineering design metric. Use NORM.INV with the overall mean (B1) and standard deviation (B2).

Col E: 85th percentile speed in mph · Col D: z-score at 85th percentile (=NORM.S.INV(0.85), 3 dec places)

B8 (3 pts) — Mean Speed: AM Peak

Compute the average speed for AM Peak observations only.

B9 (3 pts) — Standard Deviation of Speed: AM Peak

Compute the sample standard deviation of speed for AM Peak observations.

B10 (3 pts) — Mean Speed: PM Peak

Compute the average speed for PM Peak observations only.

B11 (3 pts) — Standard Deviation of Speed: PM Peak

Compute the sample standard deviation of speed for PM Peak observations.

B12 (4 pts) — 85th Percentile Speed: AM Peak

Using the AM Peak conditional mean (B8) and standard deviation (B9), compute the 85th percentile speed with NORM.INV.

B13 (4 pts) — 85th Percentile Speed: PM Peak

Using the PM Peak conditional mean (B10) and standard deviation (B11), compute the 85th percentile speed with NORM.INV.

B14 ★ (5 pts) — $P(\text{Speed} > 45 \text{ mph} \mid \text{AM Peak})$

Among AM Peak observations, what fraction of drivers exceed 45 mph? Use the AM Peak Normal model (B8, B9).

Col E: exceedance % · **Col D:** AM Peak count (=COUNTIF(C2:C501, "AM_Peak"))

B15 (4 pts) — $P(\text{Speed} > 45 \text{ mph} \mid \text{PM Peak})$

Among PM Peak observations, what fraction of drivers exceed 45 mph? Use the PM Peak Normal model (B10, B11).

Compare with B14: a real gap between the periods tells you when enforcement or signal timing would bite hardest.

B16 (MC · 2 pts) — Speed Limit Assessment

The posted speed limit is 45 mph. Based on your results for B6, B7, B14, and B15, which statement best describes what the data shows?

- (A) The limit is appropriate — a majority of drivers comply
- (B) The limit may be under-posted — the 85th percentile speed greatly exceeds 45 mph, and most drivers exceed the posted limit
- (C) The limit should be lowered — too many drivers exceed it and the data demonstrates a safety hazard
- (D) No assessment is possible without accident rate data

B17 ★ (5 pts) — $\mu_{\ln}$ for CostPerCY (Lognormal Parameter)

The lognormal distribution is parameterized by the mean and standard deviation of the **log-transformed** data. Use your LN_Cost helper column (Col G) to compute $\mu_{\ln}$.

Col E: $\mu_{\ln}$ · **Col D:** cost record count (=COUNTA(F2:F501)), should equal 500)

$\mu_{\ln}$ and $\sigma_{\ln}$ are log-units, not dollars — LOGNORM.DIST converts back.

B18 (5 pts) — P(CostPerCY > \$250) Using the Lognormal Model

First compute $\sigma_{\ln}$ in Work!B13, the *Question B18 — Lognormal spread* block: the standard deviation of your LN_Cost column. Then compute the lognormal exceedance probability for \$250 using LOGNORM.DIST with $\mu_{\ln}$ from B17 and Work!B13.

Where the working goes. Your Answer Sheet has a **Work** sheet with a labelled, pre-filled block for each table below. Build them there and nowhere else — the charts are scored by looking at the cells they plot, so a table in a blank corner of another sheet is a table your preceptor has to hunt for.

Required Charts (9 pts)

All 3 charts belong on the **Charts sheet** of your Answer Sheet, in the outlined boxes. That sheet lists what each chart is graded on.

Build the chart wherever your data is, then right-click it -> **Move Chart** -> **Object in** -> Charts, and drag it into the matching box. Leave it as a live chart — do not replace it with a picture of itself. Your preceptor clicks the chart to see which cells it plots, which is how a correct chart built from the wrong column gets caught.

Chart 1 (4 pts) — Histogram: Concrete Compressive Strength

Build a histogram of fc_psi (D2:D501) with at least 10 bins.

- Title: Histogram of Concrete Compressive Strength (fc_psi) — n=500
- Axes: Compressive Strength (psi) on x, Frequency on y

Chart 2 (3 pts) — Histogram: Traffic Speed

Build a histogram of Speed_mph (E2:E501) with at least 10 bins.

- Title: Histogram of Traffic Spot Speeds (Speed_mph) — n=500
- Axes: Speed (mph) on x, Frequency on y

Chart 3 (2 pts) — Bar Chart: ACI 318 Failure Rate by Supplier

You built the supplier block on the Work sheet back in A7 and A9. Finish its last column — D7 and D8 — with =NORM.DIST(4000, B7, C7, TRUE)*100 and the same for row 8, then plot A7:A8 against D7:D8 as a clustered bar chart with data labels on each bar.

- Title: ACI 318 Failure Rate by Supplier — P(fc < 4,000 psi)
- Axes: Supplier on x, Failure Probability (%) on y
- Add a text annotation or horizontal marker at 10% to show the ACI 318 compliance threshold

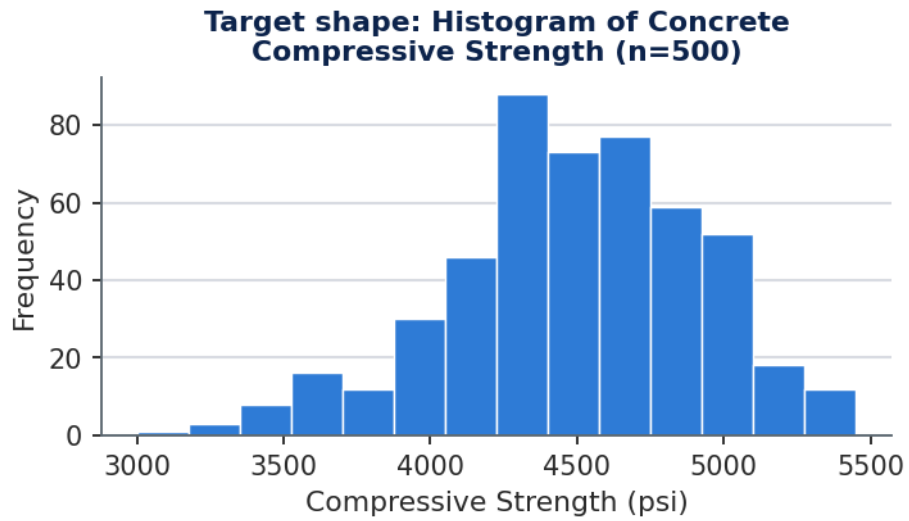


Figure 1: Target shape for Chart 1

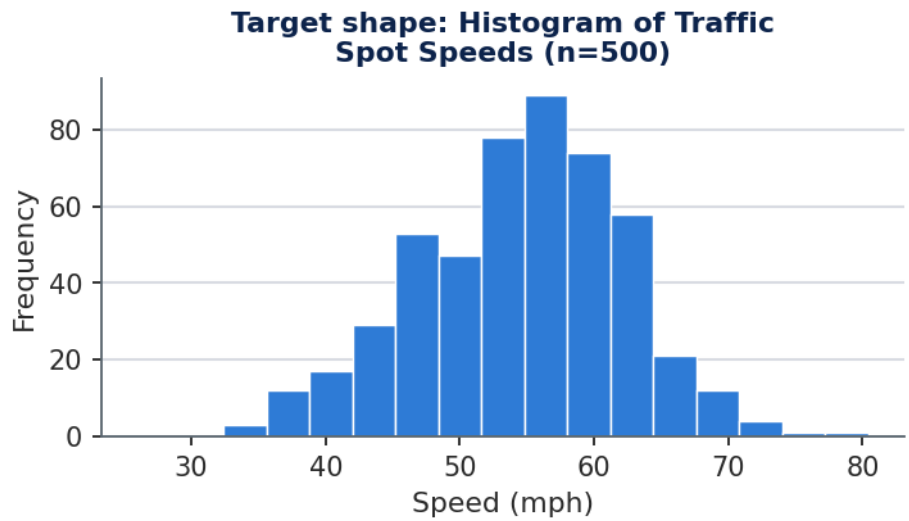


Figure 2: Target shape for Chart 2

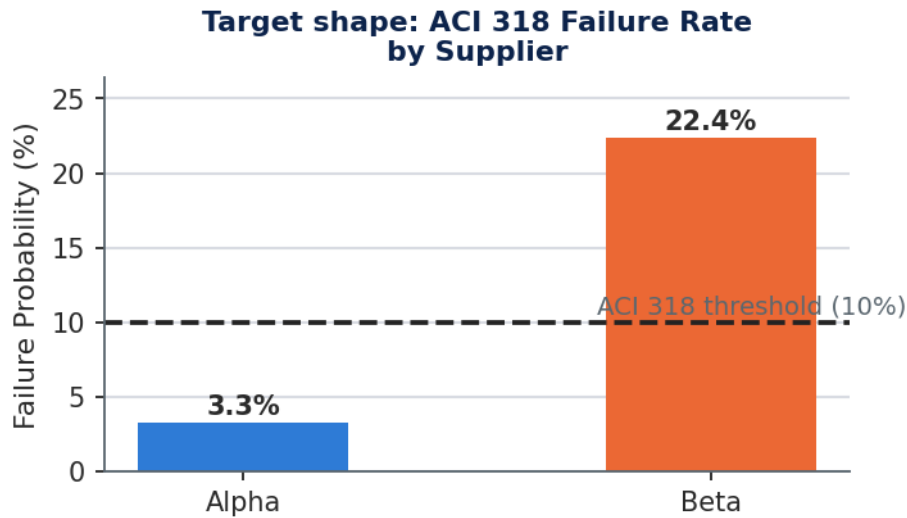


Figure 3: Target shape for Chart 3

Memo Deliverable (15 pts)

Type your memo in the **Written sheet** of your Answer Sheet Excel file. The sheet has labeled cells for each paragraph (P1, P2, P3). Press **Alt+Enter** within a cell to add line breaks.

To: CE 310 Instructor
 From: [Your Name], [Your Major]
 Date: [Date]
 Subject: Week 5 In-Class Exercise – Normal Distribution in CE Measurements

Paragraph 1 (5 pts) — Concrete Strength and ACI 318

Report the mean and standard deviation for all 500 specimens and for each supplier separately. Interpret what the difference in standard deviation between Supplier Alpha and Supplier Beta means for ACI 318 compliance. Cite the specific 10th percentile values from A7 and A9, and state which supplier complies with the 10% failure criterion and why.

Paragraph 2 (5 pts) — Traffic Speed Analysis

Report the 85th percentile speeds for the full dataset (B7), AM Peak (B12), and PM Peak (B13). Compare all three to the posted speed limit of 45 mph. Report the exceedance probabilities from B14 and B15 for each period and discuss what these results suggest about whether the posted limit reflects actual operating speeds.

Paragraph 3 (5 pts) — The Limits of Averages

A project owner says: “Our average concrete strength is well above the 4,000 psi spec, so we are fine.” Using your exercise results, explain why this reasoning is incomplete. Your answer must

reference at least two specific numbers from the in-class exercise (for example, a failure probability and a standard deviation) and identify the statistical concept the project owner is overlooking.

Submission: Save as CE310_W5_<NetID>.xlsx and upload to the Week 5 D2L dropbox. Enter your NetID in the **NetID** cell at the top of the Answers sheet — that cell is what matches your work to your student record. A 5-point deduction applies if it is blank.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet — a blank NetID cell takes a 5-point deduction
 - ☐ File saved as .xlsx and named CE310_W5_<NetID>.xlsx
 - ☐ Sheet tabs not renamed, reordered, or deleted
 - ☐ Every answer typed into its designated cell in **Col E** — answers placed anywhere else receive no credit
 - ☐ Your formula typed as text in **Col C**, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in **Col D**
 - ☐ All three charts sitting in their boxes on the Charts sheet, as live charts and not pictures, with titles and axis labels
 - ☐ Memo written in the Written sheet
 - ☐ Uploaded to D2L by **9:00 AM next Wednesday**
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Grading Summary

Component	Points
Section A — A1–A10	16
Section B — B1–B18	60
Chart 1 — Histogram: fc_psi	4
Chart 2 — Histogram: Speed_mph	3
Chart 3 — Bar chart: ACI 318 failure rate by supplier	2
Memo (3 paragraphs)	15
Total	100